

SustainaFund — Portfolio Selection

Interim review

\$100M mid-term mandate · 1,087 stocks · MIQP in FICO Xpress

What we have, why this model, how the data was cleaned, and what the numbers say

- The model is a mixed-integer quadratic program — Markowitz with cardinality, concentration and ESG constraints, solved in FICO Xpress in under a second.
- Every non-obvious choice in it was measured rather than assumed, and the measurements are reproducible from the scripts.
- One result is negative and reported as such: out of sample the optimiser and an equal-weight benchmark are statistically indistinguishable on risk-adjusted return.

The problem

Two objectives that cannot both be maximised, under six hard constraints

Constraint	Value
Budget	\$100,000,000, fully invested
Position size	1% - 20% of budget if held
Diversification	at least 30 different stocks
Region	no region above 60%
ESG, weighted	at least 70
ESG, per stock	at least 30 (added)
Sector	no sector above 30% (added)

The last two are additions to the brief, not requirements.

The trade-off

- Maximise return — the weighted sum of expected returns. But expected return is not given in the data; it has to be estimated from ten years of prices, and the choice of estimator is itself part of the work.
- Minimise risk — the variance of the portfolio, which needs the covariance between every pair of stocks. This is what captures diversification: combining stocks that do not move together lowers total risk.
- Higher return comes with higher risk, so there is no single best portfolio — only a frontier. We present three points on it for three risk appetites.

The model

Mixed-integer quadratic program, solved with FICO Xpress in 0.3-1.1 s

<i>minimise</i>	$w' S w$	portfolio variance
<i>subject to</i>	$w' \mu \geq \text{beta}$	target return, swept
	$\sum w_i = 1$	budget fully invested
	$0.01 y_i \leq w_i \leq 0.20 y_i$	1-20% if selected
	$y_i \in \{0, 1\}$	selected or not
	$\sum y_i \geq 30$	minimum holdings
	$\sum w_i \leq 0.60$ per region	
	$\sum w_i \leq 0.30$ per sector	
	$\sum \text{ESG}_i w_i \geq 70$	weighted ESG
	$\text{ESG}_i \geq 30$ if held	per-stock floor

S = covariance matrix, μ = expected returns

Binary variables are required, not a workaround: Xpress semi-continuous variables express "0 or in [1%, 20%]" natively but cannot be counted, so the 30-stock constraint still needs them. Tested - dropping the link returns 26 positions while reporting 30.

Why quadratic and not linear

- The brief offers two routes to risk: per-stock volatilities, which give a linear model, or the covariance matrix, which gives a quadratic one. We built both.
- A linear objective is the risk you would bear if every pair of stocks moved together. It is blind to diversification: swapping a stock for another of equal volatility never changes it.
- Measured: at the same expected return the linear model carries 19.2% more true risk, and it always holds exactly the 30-stock minimum - adding a 31st cannot improve its objective.
- For its own portfolio the linear objective reads 21.95% while that portfolio's actual volatility is 13.42%. The 1.64x gap is the diversification it cannot see.

How the data was cleaned

Seven decisions, each with the number that justifies it

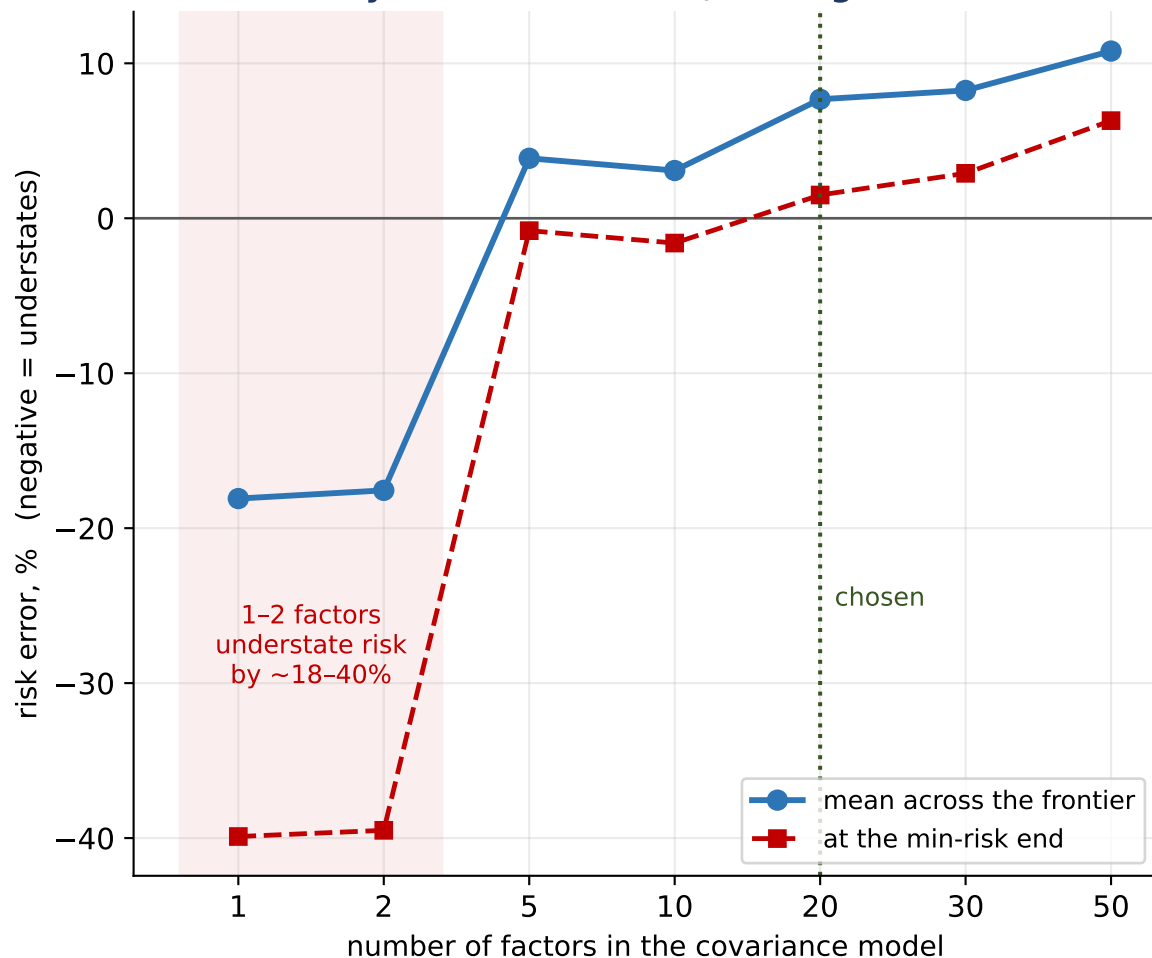
Step	Decision	Why - the measured reason
Missing ESG	49 of 1,093 filled with the 25th percentile within the region	the 25th percentile, not the mean, so a missing score is not rewarded
Price gaps	forward-filled, never before a stock's first trading day	every internal gap is exactly 1 trading day, checked on consecutive runs
Dividends	switched to total-return prices	median expected return 7.80% → 10.39%; the rise is ordered by dividend yield across sectors
Currencies	converted to USD at daily ECB rates	a local-currency Σ understated the min-variance book's risk by 19.7%
Discontinuities	ZEG.L and BMPS.MI truncated, not dropped	identical closes then a large jump = corporate action; a return across it is meaningless
Outliers	flagged, not removed	a 5σ per-stock rule fires on 0.31% of observations, mostly genuine events
Expected return	James-Stein shrinkage by sample size	a raw mean gave 88% p.a. on 460 observations; after shrinkage no stock is negative

Universe 1,093 to 1,087. Six stocks are empty in the total-return source file (three of them REITs) and were dropped rather than back-filled from unadjusted prices, which would have cost them ~3pp of return for a data defect.

Why these estimator choices

Each one changed the answer, and each was measured

Why 20 factors: a cliff, not a gradient



The three that moved the result most

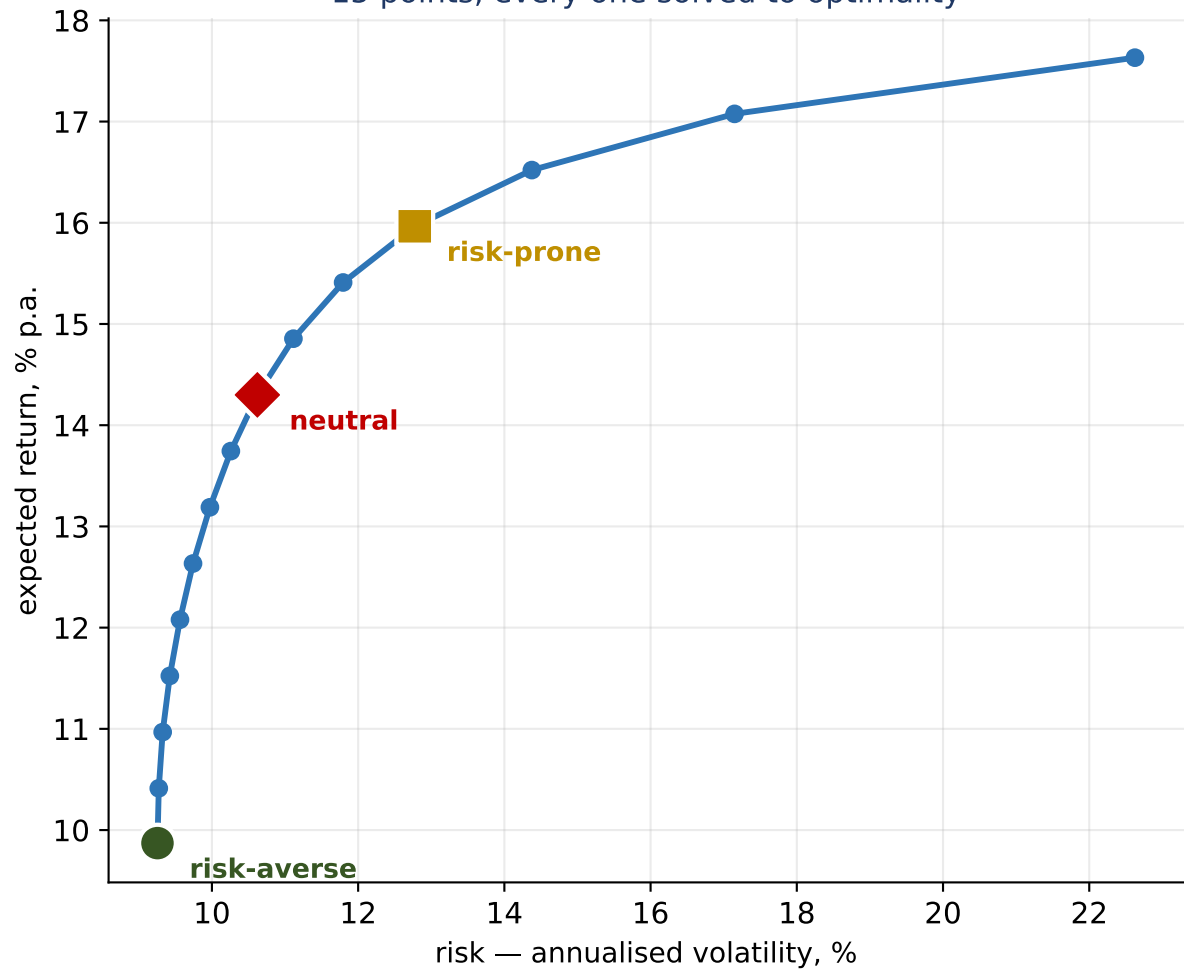
- Prices → USD. Returns are ratios, so price levels cancel — but the FX return does not: $\ln(P_{\text{usd}}) = \ln(P_{\text{local}}) + \ln(fx)$. It barely moves μ ($<0.3\text{pp}$, currencies offset) yet puts a shared factor into Σ . At the same true risk the corrected model returns 12.78% against 7.07%.
- μ shrunk. A raw historical mean is very noisy and the optimiser preferentially picks whatever that noise flattered. Shrinkage cut the frontier's top from 38.8% to 16.6% — and 38.8% would not have survived one question.
- Covariance → factor model. The complete-history rule silently excluded 96 recent listings. A factor model fits betas on whatever days a stock has, keeps all of them, and is positive semi-definite by construction.

Two things we tried that did not work, reported as such: bootstrap resampling of the weights (no effect on stability) and an explicit region factor (absorbs the structure, leaves the risk error unchanged).

Results — the efficient frontier

1,087 stocks, all constraints satisfied and verified independently of the solver

15 points, every one solved to optimality



Profile	Return	Risk	Ret/Risk	Held	ESG	Sector
risk-averse	9.87%	9.26%	1.07	42	70.0	30.0%
neutral	14.30%	10.62%	1.35	30	70.0	27.2%
risk-prone	15.97%	12.77%	1.25	30	70.0	23.0%

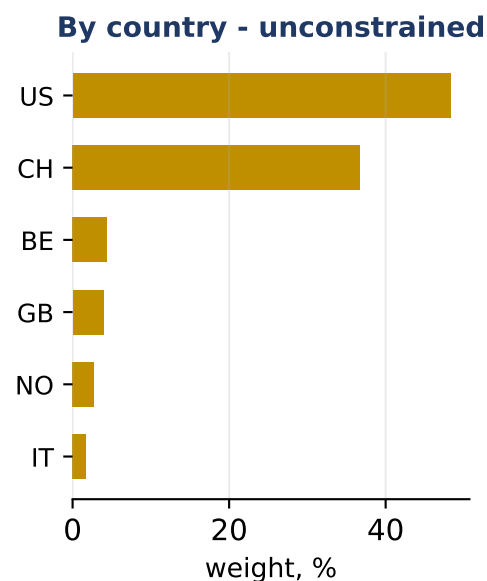
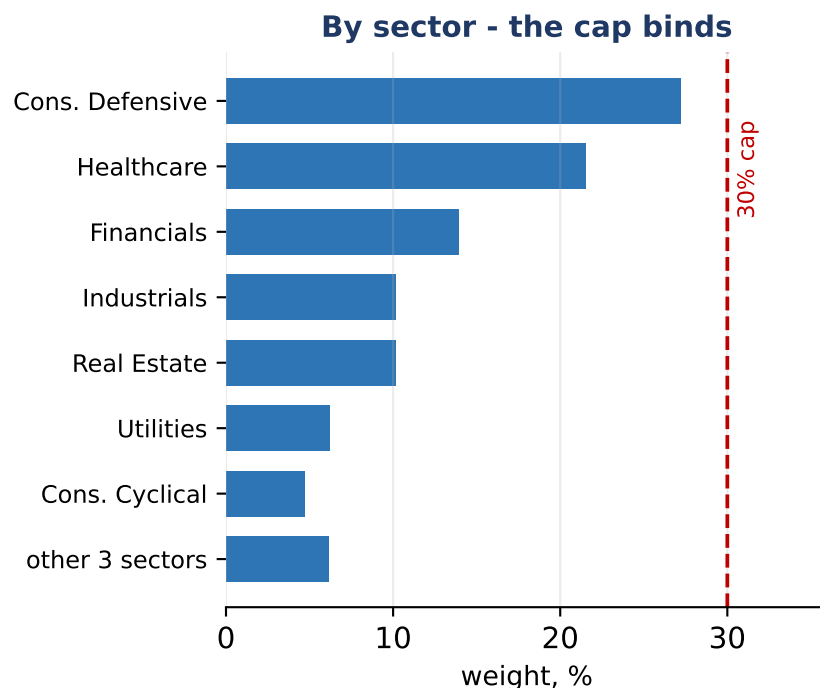
Selection is by rule, not by hand: minimum risk, maximum return/risk ratio, and the highest-return point that is not a degenerate corner.

The neutral book is the recommendation

- Best risk-adjusted profile at 1.35, 30 holdings, largest position 14.5%.
- ESG binds at exactly 70.00 in every one of the 15 points, so the constraint is always active — and costs almost nothing.
- Full holdings with dollar amounts, sector, country and ESG per position are in the results folder.

What the recommended portfolio holds

Neutral profile - 30 positions, \$100M



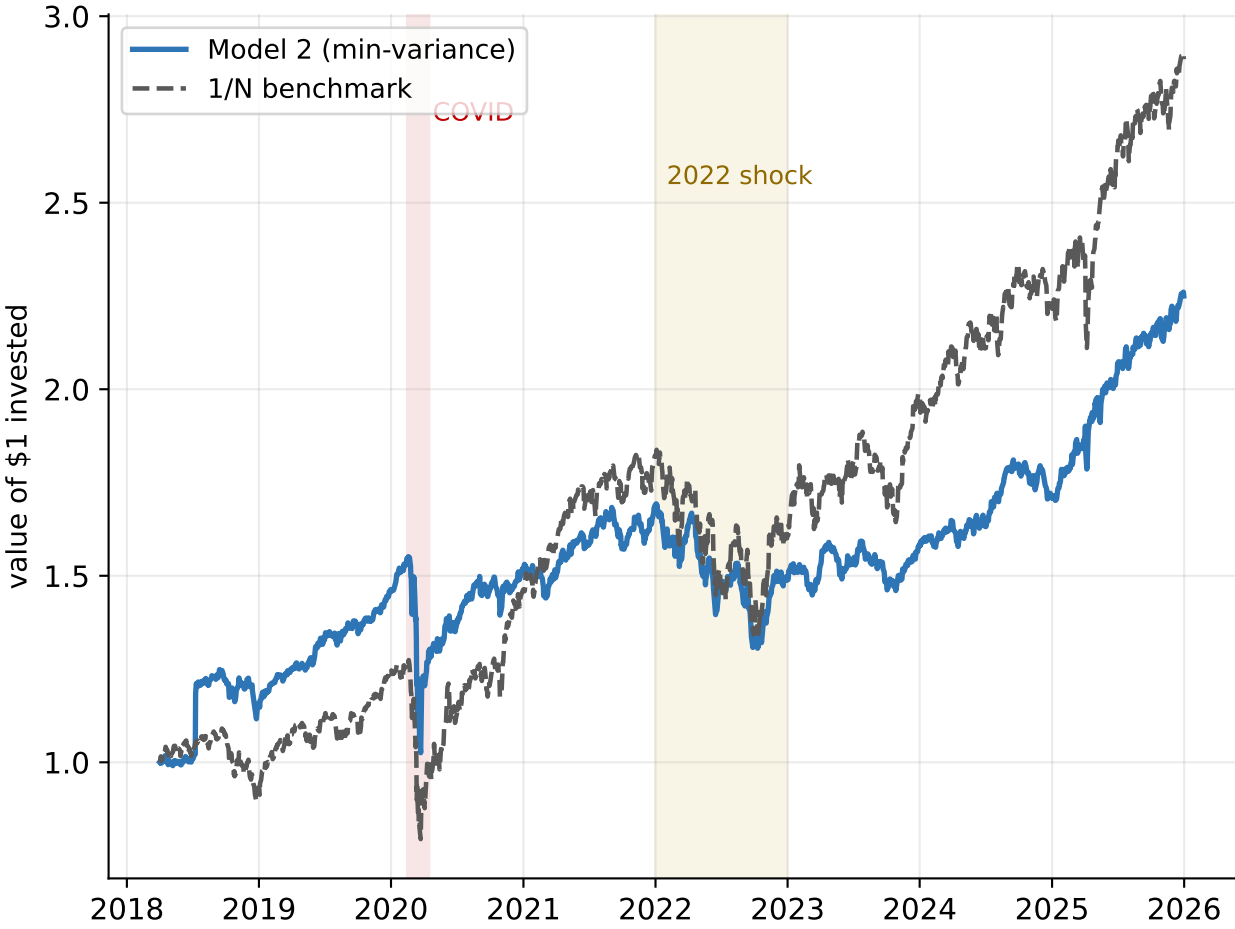
Top holdings	Wt	USD	Ctry	ESG
WMT.OQ	12.9%	\$12.9M	US	77
GALE.S	10.2%	\$10.2M	CH	69
MOBN.S	10.1%	\$10.1M	CH	82
COST.OQ	8.2%	\$8.2M	US	75
BKWB.S	6.2%	\$6.2M	CH	58
PGR.N	5.9%	\$5.9M	US	69
LLY.N	4.7%	\$4.7M	US	67
RSG.N	3.5%	\$3.5M	US	76

The concentration the sector cap cannot see

- A sector cap holds Consumer Defensive to exactly 30.00%, but nothing restrains country exposure - Switzerland reaches 47% of the risk-averse book, concentrated in cantonal banks and real estate.
- A 25% country cap held it to 25.0% for 0.156pp of return. It is not in the current model, and it is the most obvious next addition.

Walk-forward backtest

Rolling 3-year window, quarterly rebalance, 31 rebalances over 7.7 years



	CAGR	Volatility	Sharpe	Max DD
Model 2	11.03%	13.21%	0.84	-33.89%
1/N benchmark	14.65%	16.91%	0.87	-37.68%
after 10bp costs	10.85%	13.21%	0.82	-33.89%

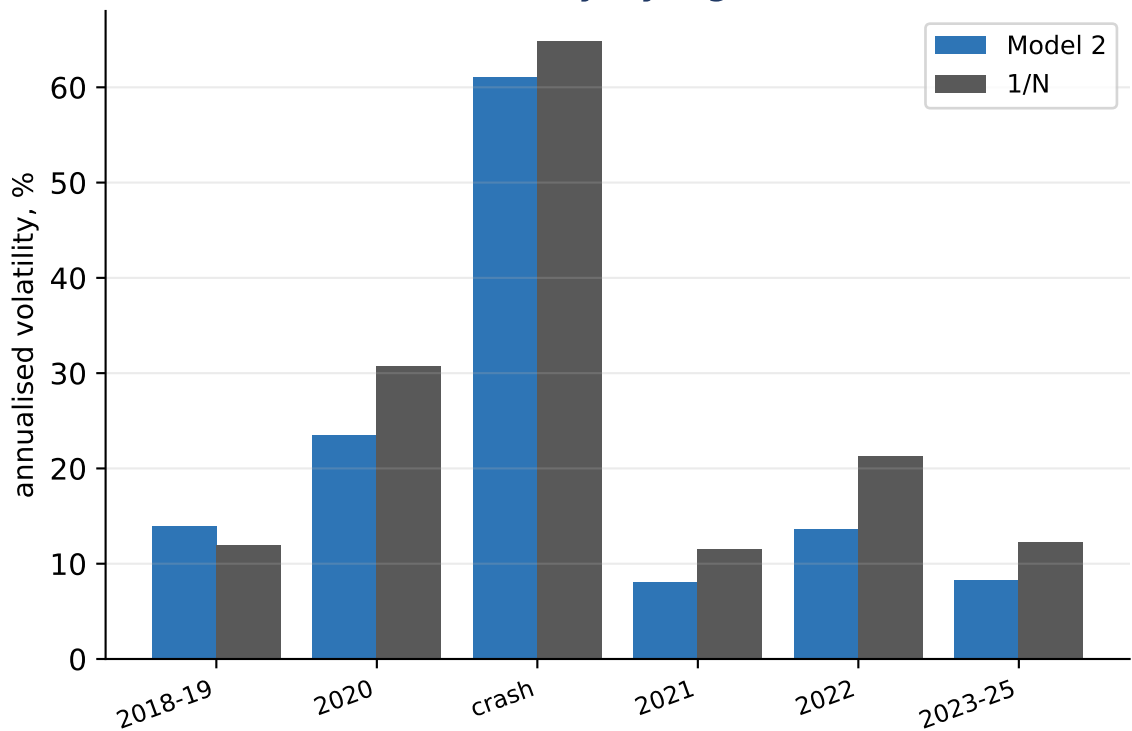
Read this honestly

- Risk-adjusted the two are statistically indistinguishable: the Sharpe gap is 0.031 and the t-statistic on the daily difference is -0.90. The claim is "indistinguishable", not "the benchmark wins".
- The volatility gap is not noise: -22%, and lower in five of six sub-periods.
- This reproduces DeMiguel, Garlappi & Uppal (2009) rather than revealing a bug. Mean-variance struggling to beat 1/N out of sample is one of the best-known results in the literature; the cause is estimation error.

Where the model earns its keep

It wins in the drawdowns and loses in the rallies — the trade a risk-averse mandate buys

Volatility, by regime



- In the COVID crash it lost 3.7pp less than the benchmark. In 2022 it fell less while running 36% lower volatility. It gives that back in every rally — 2021 and 2023-25.
- One exception, stated rather than glossed: in the calm 2018-19 window it ran higher volatility than 1/N and earned +11pp for it. Volatility is lower in five of six sub-periods, not all six.

Period	Regime	Return Δ, pp	Vol change	Max DD
2018-19	calm market	+11.1	+17%	-11% vs -18%
2020 full	crash and rebound	-14.3	-24%	-34% vs -38%
the crash	the crash itself	+6.9	-6%	-34% vs -38%
2021	boom	-13.3	-30%	-7% vs -6%
2022	inflation shock	+0.6	-36%	-23% vs -27%
2023-25	AI rally	-7.2	-33%	-8% vs -13%

Highlighted: the crash and the 2022 shock — the two periods it wins.

Limitations, measured not guessed

Four of these are properties of the data and cannot be fixed with this dataset

Limitation	The measurement	Consequence for the reader
Risk is understated out of sample	predicted 8.78% at rebalances vs 13.21% realised	say "consistent with the estimation period", never "conservative"
The estimation window is a judgement	Lockheed is picked in 100% of 5-year bootstrap resamples and 50% of 10-year ones	we chose 10 years for a full cycle; the cost is reported, not hidden
ESG and sector data are a snapshot	one present-day value per stock, no history	applying ESG ≥ 70 at a 2018 rebalance uses 2025 information — look-ahead
The universe is survivorship-biased	today's STOXX 600 and S&P 500 constituents only	returns are optimistic for every strategy here, benchmark included
Country concentration is unconstrained	Switzerland reaches 47% of the risk-averse book	a 25% country cap costs 0.156pp — the obvious next addition
The 30-stock floor pads the book	11 of 30 positions sit exactly on the 1% minimum	a higher minimum weight would give a cleaner portfolio — worth raising with the client
Transaction costs are outside the model	turnover 21% per rebalance, ~84% a year	at 10bp that is 0.18pp of CAGR, so costs are not the binding issue

The first two we can act on. The third and fourth are properties of the dataset. The last three are choices open to the client.

Where this stands, and what is next

Settled

- Model formulation, solved to optimality in under a second on 1,087 stocks
- Data cleaning — every decision measured and reproducible from a script
- Both risk models compared: quadratic beats linear by 19.2% of risk
- Three portfolios, all seven constraints verified independently of the solver
- Walk-forward backtest, 7.7 years and 31 rebalances

Open

- Country cap. Switzerland at 47% is the one concentration nothing restrains. Measured cost 0.156pp.
- Dashboard. Not started, and the largest remaining piece of work.
- Six missing series. Three REITs among them — worth re-extracting.

The one slide to remember

- The optimiser reliably delivers what it optimises — 22% less volatility than equal weight, and smaller drawdowns in every crisis in the sample.
- It does not deliver a higher risk-adjusted return out of sample. That is a known result in the literature, not a defect in this implementation, and the cause — estimation error — we measured three independent ways.
- So this should be presented as a risk-control product, not a return-maximising one. That is what the constraints describe and what the backtest supports.

*Educational exercise. Not investment advice.
Past performance does not guarantee future results.*